



THENETH SANJUKA ARIYARATHNE

Systems Security Engineer | Software Engineer | Cybersecurity
Specialist

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 Malabe, Sri Lanka

 Languages: English, Sinhala

PROFESSIONAL SUMMARY

Results-driven Systems Security Engineer and Software Engineer with rapid career progression from Junior to Software Engineer at Aurasync Technologies, alongside enterprise defense experience at Sysflicx IT Solutions and Avian Technologies. Combines deep expertise in scalable software development, DevSecOps, and continuous integration / continuous delivery (CI/CD) with infrastructure hardening, cloud security (AWS / GCP Hardening), and communication protocol defense (RF, IP/Ethernet, AES/TLS). Proven track record in embedding static/dynamic application security testing (SAST/DAST) into deployment workflows, managing enterprise firewalls (Sophos, Cyberoam), and conducting full-scope vulnerability assessment and penetration testing (VAPT) aligned with National Institute of Standards and Technology Cybersecurity Framework (NIST CSF) and ISO/IEC 27001 standards. Dedicated to delivering tamper-resilient, mission-critical systems security and safeguarding core enterprise infrastructure.

SKILLS & TECHNICAL EXPERTISE

Systems & Infrastructure Security: Cloud Security (AWS/GCP Hardening), SCADA & Industrial Control Security, Network Architecture Hardening, Firewall Configuration (Sophos, Cyberoam), VPN & Secure Gateway Setup, Endpoint Protection, Tamper-Resilient System Architecture.

Secure Communication & Cryptography: RF & IP/Ethernet Communication Protocols, AES/TLS Encryption Standards, Applied Cryptography, Wireless Network Defense, Zero-Trust Protocol Auditing.

Secure Software Engineering & DevSecOps: Scalable Software Architecture, Secure API Design (OAuth 2.0/JWT), Microservices Security, CI/CD Pipeline Integration, Code Review & Defect Remediation, Modern Web & Mobile Development.

Vulnerability Management & VAPT: Full-Scope Penetration Testing, Threat Modeling & Risk Mitigation, Incident Response & Triage, Log & Packet Flow Anomaly Analysis, Security Compliance Auditing.

Security Tools & Platforms: Metasploit, Nessus, Burp Suite, Kaspersky Security Center, McAfee, Wireshark, SIEM & Log Monitoring suites, Git Version Control.

Programming & Automation: Python (Advanced for NIDS, Automation, & ML Anomaly Detection), Java, C++, JavaScript/React, Node.js, Bash/Shell Scripting.

Compliance & Governance: NIST Cybersecurity Framework (CSF), ISO/IEC 27001, HIPAA, OCTAVE Risk Assessment Methodology, Professional Engineering Ethics & Verifiable Reporting.

PROFESSIONAL EXPERIENCE

Security Software Developer / Engineer | Aurasync Technologies | Rajagiriya, Sri Lanka

📅 July 2025 – Present

- Build custom software tools, security applications, and features designed to protect systems from threats, embedding DevSecOps practices across the SDLC to reduce post-release defects by 35% across 5+ client applications.
- Architect secure cloud infrastructure and integrate continuous vulnerability scanning (SAST/DAST) into CI/CD pipelines, accelerating deployment cycles by 25% while maintaining strict data privacy compliance.
- Provide strategic IT consulting and client advisory on application security hardening, encrypted communications (AES/TLS), and tamper-resistant software design for 10+ commercial partners, enhancing infrastructure compliance readiness by 40%.
- Mentor a junior engineering cohort in clean code architecture, API security protocols, and proactive threat mitigation.

Associate Security Software Developer / Engineer | Aurasync Technologies | Rajagiriya, Sri Lanka

📅 December 2024 – June 2025

- Designed and developed scalable web and mobile applications using Python, Node.js, and React, securing 100K+ daily user interactions against OWASP Top 10 vulnerabilities with zero breach incidents.
- Collaborated with cross-functional engineering teams to implement secure OAuth 2.0/JWT authentication and encrypted microservices data transmission across 5+ distributed enterprise environments, maintaining 99.9% system uptime.
- Performed rigorous peer code reviews, vulnerability triage, and system remediation, identifying and resolving over 40 potential software defects early in the development lifecycle.
- Contributed to custom software delivery and strategic technical consulting, aligning client digital transformation initiatives with industry security standards.

Junior Software Engineer | Aurasync Technologies | Rajagiriya, Sri Lanka

📅 May 2024 – November 2024

- Built responsive frontend interfaces and secure backend modules for commercial applications and academic support projects, achieving 99.8% system uptime and adherence to clean coding guidelines.
- Optimized database performance, integrated RESTful APIs, and remediated network communication anomalies across 15+ multi-server staging and production environments, reducing latency by 20%.
- Authored comprehensive technical documentation, system architecture specifications, and client requirement analyses to streamline digital solutions delivery.

Associate Information Security Analyst | Sysflicx IT Solutions | Pannipitiya, Sri Lanka

📅 June 2023 – May 2024

- Spearheaded incident response support and threat triage across diverse enterprise client environments, analyzing 30+ weekly security alerts and reducing Mean Time to Remediation (MTTR) by 25%.
- Conducted comprehensive network vulnerability assessments and security audits for 10+ corporate infrastructures, achieving 100% alignment with NIST CSF and ISO 27001 compliance standards.
- Collaborated directly with systems engineering and network teams to harden defense mechanisms, deploy security patches across 200+ endpoints, and optimize perimeter monitoring.
- Authored technical risk assessments and security policy documentation, conducting organizational outreach that elevated staff phishing awareness and compliance by over 40%.

Network Security Analyst Intern | Avian Technologies | Nawala, Sri Lanka

 **November 2022 – May 2023**

- Configured, deployed, and maintained enterprise network firewalls (Sophos, Cyberoam) and perimeter security appliances across 15+ corporate deployments to prevent unauthorized network access.
- Managed centralized antivirus and endpoint defense suites across 100+ endpoints, ensuring 100% signature compliance, and implemented encrypted site-to-site VPN tunnels for secure remote access.
- Monitored real-time network traffic and system logs for suspicious behaviors using Wireshark and SIEM tools, conducting periodic vulnerability scans to ensure infrastructure resilience.
- Assisted senior engineers in troubleshooting RF and IP/Ethernet network communication protocols and implementing encrypted site-to-site VPN tunnels for secure remote enterprise access.

Machine Learning & Systems Security Specialist (Freelance) | Self-Employed

 **January 2020 – Present**

- *Scope:* Delivering custom infrastructure defense, security automation, and penetration testing for commercial and enterprise client networks.
- Engineered 15+ custom Python automation scripts and specialized vulnerability scanning tools to systematically monitor infrastructure integrity across 10+ client networks, reducing manual detection time by 60% and identifying zero-day system anomalies with 99.4% accuracy.
- Conducted full-scope network and web application penetration tests using Metasploit, Nessus, and Burp Suite, identifying and remediating 50+ critical and high-severity vulnerabilities before client production deployments.
- Delivered executive-level remediation reports, system architecture hardening blueprints, and tailored risk mitigation strategies for commercial clients, ensuring 100% audit pass rates.
- Researched and integrated Machine Learning algorithms to enhance anomaly detection accuracy by 30% in high-volume network traffic and system log analysis.

RESEARCH & PUBLICATIONS

Developing an Optimal Strategy to Address the Vulnerability of Image Tampering

International Research Journal of Innovations in Engineering and Technology (IRJIET) | ISSN (Online): 2581-3048, Volume 7, Issue 11, November 2023, pp. 511–519

- Authored peer-reviewed research focusing on forensic integrity, digital anomaly detection, and optimal algorithmic mitigation strategies against data tampering.

 [View Published Research Paper](#)

Enhancing Organizational Time Efficiency Using Machine Learning for Employee Activity Monitoring

International Journal of Scientific and Research Publications (IJSRP) | ISSN: 2250-3153, Volume 14, Issue 9, September 2024 | DOI: 10.29322/IJSRP.14.09.2024.p15309

- Co-authored peer-reviewed research investigating real-time computer vision (CCTV footage analysis) and facial recognition algorithms to automate employee activity monitoring and optimize organizational time efficiency.

 [View Published Research Paper on ResearchGate](#)

KEY PROJECTS & TECHNICAL HIGHLIGHTS

- **Enterprise DevSecOps CI/CD Pipeline (August 2025 – November 2025 | Aurasync Technologies):** Integrated streamlined SAST/DAST vulnerability scanners into cloud deployment workflows, reducing post-release defects by 35%.
- **Secure Microservices OAuth 2.0 Gateway (February 2025 – May 2025 | Aurasync Technologies):** Architected an encrypted authentication gateway handling 100K+ daily API transactions with zero security breaches.
- **Commercial Web & Mobile Portal Hardening (June 2024 – September 2024 | Aurasync Technologies):** Developed responsive React/Node.js modules and remediated OWASP Top 10 vulnerabilities for digital transformation clients.
- **Multi-Site Corporate VAPT & Compliance Audit (January 2024 – April 2024 | Sysflicx IT Solutions):** Carried out full-scope penetration testing across 5+ enterprise networks using Metasploit and Burp Suite, achieving ISO 27001 alignment.
- **Systematized SIEM Triage & Alert Optimization (July 2023 – October 2023 | Sysflicx IT Solutions):** Engineered custom log correlation rules to triage 30+ weekly security alerts, reducing Mean Time to Remediation (MTTR) by 25%.
- **Multi-Site Firewall & IPS Deployment (February 2023 – April 2023 | Avian Technologies):** Configured and hardened Sophos and Cyberoam perimeter firewalls across 15+ corporate deployments with active intrusion prevention.
- **Encrypted Site-to-Site VPN Architecture (December 2022 – January 2023 | Avian Technologies):** Implemented AES/TLS-encrypted VPN tunnels to establish tamper-resilient remote enterprise access and communication redundancy.
- **Next-Gen NIDS & Threat Intelligence Engine (March 2024 – June 2024 | Freelance):** Developed a custom Python/ML Network Intrusion Detection System to inspect real-time packet flows and block zero-day anomalies.
- **SCADA & IoT Communication Protocol Scanner (August 2022 – November 2022 | Freelance):** Built a custom Python vulnerability scanner to audit RF and IP/Ethernet command packets for industrial control environments.

CERTIFICATIONS & INDUSTRY CREDENTIALS

- **ISC2 Certified in Cybersecurity (CC)** — *International Information System Security Certification Consortium (ISC2)*
- **CISA Cybersecurity for Industrial Control Systems (ICS-210w)** — *U.S. Cybersecurity and Infrastructure Security Agency (DHS)*
- **Fortinet Certified Associate (FCA) in Cybersecurity** — *Fortinet Training Institute*
- **Qualys Certified Specialist in Vulnerability Management (VMDR)** — *Qualys Enterprise Security*
- **Cisco Junior Cybersecurity Analyst Career Path** — *Cisco Networking Academy*

AWARDS & RECOGNITION

- **Outstanding Engineering Excellence Award (2025) — Aurasync Technologies:** Recognized by executive leadership for spearheading secure cloud architecture and integrating programmatic DevSecOps vulnerability scanning across client digital transformation initiatives, resulting in zero post-release security breaches.

EDUCATION

BSc (Hons) in Information Technology, Specializing in Cybersecurity

Kingston University, UK (Graduated: 2026)

Higher National Diploma (HND) in Information Technology, Specializing in Cybersecurity

Sri Lanka Institute of Information Technology (SLIIT) (Graduated: 2023)

G.C.E. Advanced Level (A/L) Examination (2018)

C.W.W. Kannangara Central College, Mathugama | Results: 1 B & 2 Cs (Biological Science Stream)

G.C.E. Ordinary Level (O/L) Examination (2014)

C.W.W. Kannangara Central College, Mathugama | Results: 8 As & 1 B

PROFESSIONAL REFERENCES

Dr. Lakmal Rupasinghe

Senior Lecturer, Department of Information Systems Engineering, Faculty of Computing – SLIIT

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Mr. Eranga Jayapathma

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DECLARATION

I hereby affirm that the information provided in this document is accurate and complete to the best of my knowledge. I am committed to upholding the highest standards of professionalism and integrity in my work. If selected, I am dedicated to contributing effectively to the team and organization, utilizing my skills and knowledge to their fullest potential.

Date: 25th July 2026

Place: Malabe, Sri Lanka



Theneth Sanjuka Ariyaratne